



Penetrating head & neck trauma – Epidemiology and injury characteristics in terror-related violence, interpersonal violence and deliberate self-harm at a level 1 trauma centre

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ABSTRACT

Introduction: Penetrating trauma to the head and neck presents specific clinical challenges. Aetiologies include interpersonal violence, deliberate self-harm and terror-related violence. King's College Hospital is a Major Trauma Centre serving inner-city London boroughs with a high incidence of knife and gun crime. It also received victims of the terrorist attack at London Bridge in June 2017.

Methods: Data was collected prospectively on all patients presenting with penetrating trauma to the head and neck over a one-year period (August 2016–July 2017).

Results: Overall figures for penetrating trauma are the highest since comparable records began with 478 cases (2016/17) compared with 172 (2010/11). Most patients had injuries resulting from interpersonal violence (83%); a group consisting mostly in males (88%) under the age of 30 (69%). The sole fatality among all patients was a result of repeat deliberate self-harm. Terror-related violence victims were equally gender split and older; all patients in this group required surgical intervention and had longer lengths of stay (16.3 days) compared with interpersonal violence (6.3 days) and deliberate self-harm (3 days).

Conclusion: Violent crime is increasing. Overall penetrating injury has more than doubled in 6 years. The injury characteristics observed are distinct depending on the assailants and motives. Head and neck trauma requires a multidisciplinary approach, with an understanding of the care needs of patients. Interpersonal violence is the most common aetiology and as such efforts to tackle rising violent crime must involve police and the community. In addition, terror-related violence is evolving in the United Kingdom and globally, and as such healthcare professionals must understand differences in aetiologies to promote optimal patient care.

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Introduction

King's College Hospital is a Major Trauma Centre serving an inner-city population of around 700,000, while acting as a tertiary referral centre for many millions across the south of

England. The hospital straddles two London boroughs with some of the highest incidence of knife crime in the country.¹ King's became a Major Trauma Centre in 2010 following recommendations regarding trauma care made in the Darzi review. As a result, it became the hub for a network of trauma 'units' across southeast London and Kent, stretching from

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